ANALYSIS OF PUBLIC SENTIMENT ON FLUE VACCINES

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Introduction

Public opinions about flu vaccines vary widely across different platforms. While many people support vaccination, others express concerns about safety, side effects, and trust.

In this project, we analyze large amounts of online data from social media, news, and medical reports to better understand how people feel about flu vaccines.

Our goal is to identify patterns in public sentiment and provide insights that can support better health communication and awareness.

Objectives

1. Understand overall public sentiment (positive, neutral, negative)
2. Identify common concerns and opinions about flu vaccines
3. Compare sentiment across different sources (social media, news, medical data)
4. Analyze how public opinion changes over time

Data Collection

We collected data from multiple sources to capture different perspectives:

- **YouTube** – Public comments and discussions
- **News articles (GNews)** – Media coverage of vaccines
- **OpenFDA** – Official medical reports on vaccine reactions

Combining these sources helps us understand both public opinion and real-world medical data.

Data Analysis

Our analysis revealed key patterns:

- Social media discussions often include concerns and **negative** opinions
- News articles provide **balanced** and informational content
- Medical data highlights common reported reactions
- Public sentiment **varies** depending on the source

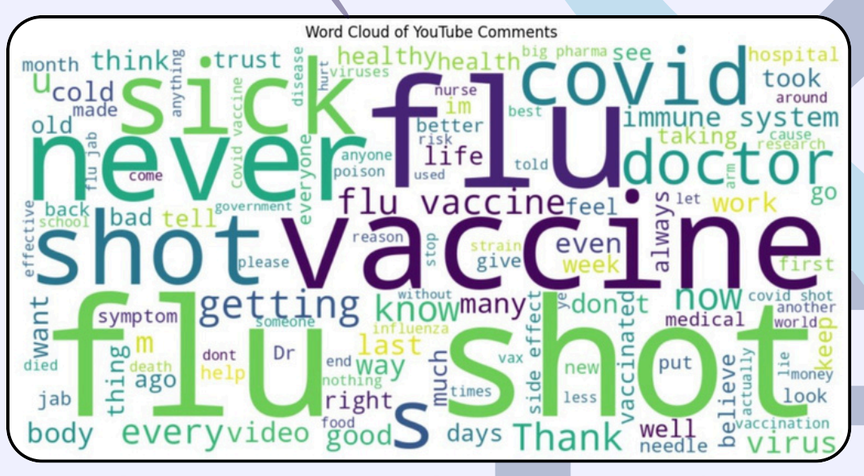


Figure 1 — Most Common Words

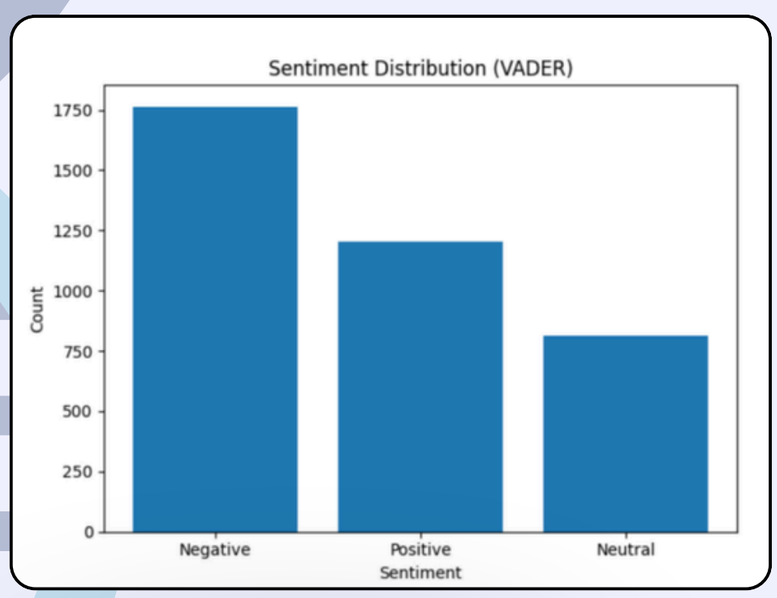


Figure 2 — Sentiment Distribution of Comments

Results and Findings

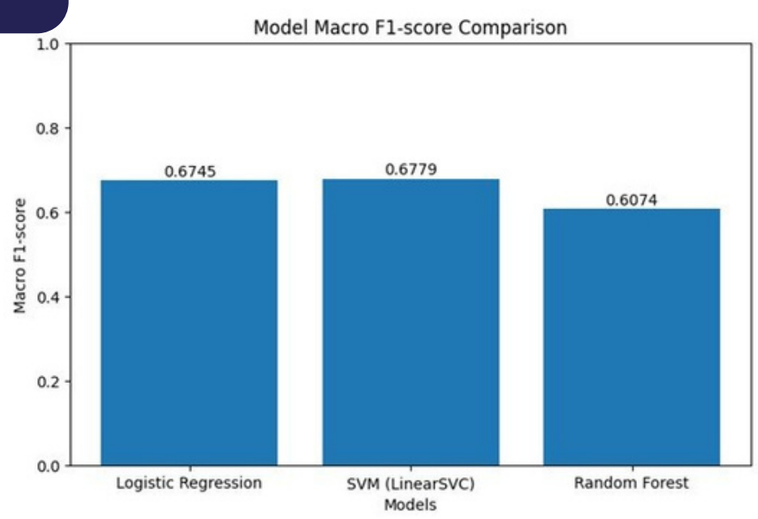
We tested three machine learning models to classify sentiment:

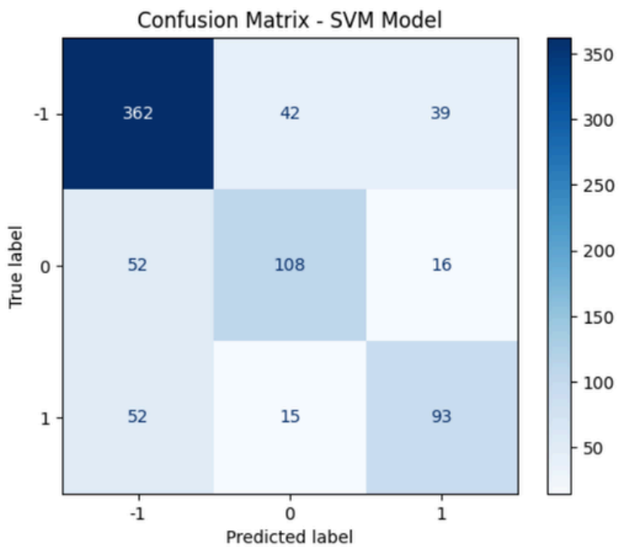
- **Logistic Regression (baseline)**
- **Support Vector Machine (SVM)**
- **Random Forest**

Since the dataset is imbalanced, accuracy alone is not reliable. Therefore, we used Macro F1-score, which gives equal importance to all sentiment classes.

Results Showed;

- **SVM** achieved the best overall performance
- All models **performed well on negative** sentiment
- Neutral and positive sentiments were harder to classify





Conclusion

- This project shows that public opinions about flu vaccines are complex and vary across platforms.
- Social media often contains more negative or emotional discussions, while news and medical data provide more structured insights.
- By combining multiple data sources and using machine learning, we were able to better understand public sentiment.
- The SVM model provided the most balanced and reliable performance.
- These insights can help improve public health communication and address concerns about vaccines.